

6-mavzu. Generativ modellar: autoencoder, VAE va GAN

2026–2027 o'quv yili · 30 ta topshiriq

Muhitni sozlash. Barcha kodlar Python 3.11/3.12 (CPU) va quyidagi qat'iy versiyalar bilan sinovdan o'tgan (requirements.txt); ma'lumotlar internetsiz: sklearn.datasets.load_digits (8×8 , paket ichida), make_moons va sintetik Gauss.

```
python -m venv .venv && source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements.txt
# numpy==2.4.4 scipy==1.17.1 pandas==3.0.2 scikit-learn==1.8.0 scikit-image==0.26.0
# matplotlib==3.10.9 pillow==12.2.0 opencv-python-headless==4.13.0.92 torch==2.14.0 torchvision==0.29.0
```

Hujjatlar: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_digits.html, <https://docs.pytorch.org/docs/stable/distributions.html>, <https://docs.pytorch.org/docs/stable/generated/torch.nn.BCEWithLogitsLoss.html>, <https://docs.scipy.org/doc/scipy/reference/generated/scipy.linalg.sqrtm.html>.

Mavzu. Autoencoder va PCA, denoising va Tweedie formulasi, KL divergence, VAE (ELBO, reparameterization trick, β -VAE, posterior collapse), GAN (optimal diskriminator, JSD, non-saturating loss, mode collapse), normalizing flows, baholash (FID, PSNR).

Maqsadlar. KL, ELBO, D^* va JSD ni qo'lda hisoblash va isbotlash; VAE yo'qotishini to'g'ri masshtabda yozish; load_digits da autoencoder, VAE va GAN ni o'qitib, tiklash, KL, sinflar taqsimoti va FID bilan taqqoslash; anomaliyani aniqlash; kichik RealNVP flow bilan aniqlik $p(\mathbf{x})$ ni o'lchash.

Baholash (100 ball). 1–16: har biri 2 ball (32); 17–28: har biri 4 ball (48); 29, 30: har biri 10 ball (20). Kod topshiriqlarida: to'g'ri ishlashi 60%, talab qilingan chiqish formati 20%, natija talqini 20%. Umumiy talab: har bir kod topshirig'i alohida funksiya; torch.manual_seed(42), np.random.default_rng(42), CPU; sonlar 4 xonagacha.

A. Qo'lda hisoblash, isbot va tushunchalar (1–16)

- Chiziqli autoencoder va PCA.** Markazlashgan ma'lumot $(4, 2), (-4, -2), (1, -2), (-1, 2)$; $k = 1$. $C = \frac{1}{4}X^TX$, xos qiymatlar va birinchi bosh yo'nalish $\mathbf{u}$; optimal chiziqli AE tiklashlari va o'rtacha kvadratik xato.
- Tweedie.** $x \sim \mathcal{N}(2, 1)$, $\tilde{x} = x + \sigma\epsilon$, $\sigma^2 = 0,5$. $\mathbb{E}[x | \tilde{x}]$ ni to'g'ridan-to'g'ri va $\tilde{x} + \sigma^2 \nabla \log p_\sigma(\tilde{x})$ formulasi bilan toping; $\tilde{x} = 3,5$ da qiymat.
- KL assimetriyasi.** $P = (0,2; 0,8)$, $Q = (0,5; 0,5)$: $\text{KL}(P\|Q)$ va $\text{KL}(Q\|P)$ (nat).
- Gauss KL.** $q = \mathcal{N}(\boldsymbol{\mu}, \text{diag } \boldsymbol{\sigma}^2)$, $\boldsymbol{\mu} = (1; -0,5)$, $\boldsymbol{\sigma} = (0,5; 2)$: $\text{KL}(q\|\mathcal{N}(\mathbf{0}, I))$ (har koordinata alohida).
- Isbot.** a) Gibbs tengsizligi $\text{KL}(q\|p) \geq 0$. b) $\text{KL}(\mathcal{N}(\mu_1, \sigma_1^2)\|\mathcal{N}(\mu_2, \sigma_2^2)) = \ln \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$; $\mathcal{N}(1, 1)$ va $\mathcal{N}(0, 4)$ uchun hisoblang.
- ELBO.** ELBO ni Jensen tengsizligi orqali chiqaring va bo'shliq $\text{KL}(q\|p(\mathbf{z}|\mathbf{x}))$ ekanini ko'rsating. Model: $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, 1)$, $p(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{z}, 1)$, $x = 1$. $\log p(\mathbf{x})$, posterior, $q = \mathcal{N}(\mathbf{0}, 1)$ uchun ELBO va bo'shliq.
- VAE yo'qotishi.** $\mathbf{x} = (0; 1; 1)$, $k = 1$, $\mu = -0,5$, $\sigma^2 = 0,25$, $\varepsilon = 0,4$; Bernulli decoder logit = $\mathbf{w}\mathbf{z} + \mathbf{b}$, $\mathbf{w} = (1; -2; -3)$, $\mathbf{b} = (0; 0,5; 0)$. z , $\hat{\mathbf{x}}$, BCE (yig'indi), KL va L .
- Reparameterization.** 7-topshiriqda $\partial z / \partial \mu$, $\partial z / \partial \sigma$, $\partial z / \partial \log \sigma^2$ ni toping. Nega torch.normal(mu, std) dan olingan namuna bilan encoder o'qitilmaydi?
- Masshtab va β .** 8×8 rasm uchun BCE reduction='mean' bilan yozilsa, amalda β nechaga teng bo'ladi? $\beta \rightarrow \infty$ da optimal encoder va decoder nimaga intiladi? Posterior collapse nima?
- Optimal diskriminator.** $p_{\text{data}} = \mathcal{N}(0, 1)$, $p_g = \mathcal{N}(2, 1)$: $D^*(x)$ ni logistik ko'rinishda yozing; $D^*(0)$, $D^*(1)$, $D^*(2)$.
- JSD.** $C(G) = -\log 4 + 2 \text{JSD}(p_{\text{data}}\|p_g)$ ni isbotlang. $P = (0,5; 0,5)$, $Q = (0,8; 0,2)$ uchun M , JSD va C .

12. **Non-saturating loss.** $D(G(z)) = 0,05$ da $\log(1 - \sigma(a))$ va $-\log \sigma(a)$ ning logit a bo'yicha hosilalari; nisbati. Muvozanatda ($D \equiv \frac{1}{2}$) L_D va L_G qiymatlari.
13. **Minimax tebranishi.** $f(x, y) = xy$, $(x_0, y_0) = (2; 0)$, $\eta = 0,2$, bir vaqtdagi qadamlar. Har qadamda norma necha marta o'sishini isbotlang; 10 qadamdan keyingi norma.
14. **Flow.** $z \sim \mathcal{N}(0, 1)$, $x = 3z - 1$: change of variables bilan $\log p_X(2)$ ni toping va $\mathcal{N}(-1, 9)$ zichligi bilan tekshiring. Affine coupling qatlamining yakobian determinanti nega arzon hisoblanadi?
15. **Baholash.** a) Bir o'lchovli FID: $\mathcal{N}(0, 1)$ va $\mathcal{N}(0,5; 0,5^2)$. b) $\mu_r = (0; 0)$, $\Sigma_r = I$, $\mu_g = (1; 0)$, $\Sigma_g = \text{diag}(4; 0,25)$: FID. c) $[0, 1]$ piksellarda MSE = 0,005: PSNR.
16. **Tushunchalar.** a) Autoencoder, VAE, GAN ni maqsad, yashirin fazo va zichlik bo'yicha solishtiring. b) Nega GAN yo'qotishi sifat ko'rsatkichi emas? c) `fake.detach()` nima uchun kerak? d) $p_{\text{data}} = \delta_0$, $p_g = \delta_\theta$ misolida JSD va Wasserstein masofasini solishtiring; mode collapse qanday aniqlanadi?

B. Dasturlash (17–30)

Har bir funksiya natijani [T<nomer>] prefiksi bilan chop etadi; hammasini `main()` ishga tushiradi. Ma'lumot (19–25, 29): `X, y = load_digits(return_X_y=True)`; `X = (X/16).astype(np.float32)`; `Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.25, random_state=42, stratify=y)` (1347/450).

17. **KL va reparameterization.** `kl_gauss(mu, logvar)` (oxirgi o'q bo'yicha yig'indi); 4-topshiriqni tekshiring. `torch.manual_seed(42)`; $\mu = 1$, $\sigma = 2$ (`requires_grad`), `eps=torch.randn(400000)`, $z = \mu + \sigma \epsilon$; `torch.distributions.Normal` bilan Monte-Karlo KL; so'ng `(z**2).mean().backward()`: $\partial/\partial\mu$ va $\partial/\partial\sigma$ ni nazariy 2μ , 2σ bilan solishtiring.
18. **VAE yo'qotishi.** `vae_loss(x, logits, mu, logvar, beta=1.0, reduction='sum')` -> (L, rec, kl) (`binary_cross_entropy_with_logits`). 7-topshiriqni tekshiring; 'mean' bilan KL/BCE nisbati qanday o'zgaradi?
19. **Chiziqli AE = PCA.** Train o'rtachasi bilan markazlang; `nn.Linear(64,8,bias=False)` va `nn.Linear(8,64,bias=False)`; `torch.manual_seed(42)`; Adam $1e-2$, 3000 to'liq-paket qadam, MSE. Test MSE ni PCA(8) bilan; $W_2 W_1$ ni $U U^T$ bilan (nisbiy Frobenius farq), qism-fazolar orasidagi burchak kosinuslari (`qr + svd`), dispersiyasi 0 bo'lgan piksellar soni, ma'lumotda $\max |P_{ae} \mathbf{x} - P_{pca} \mathbf{x}|$ va λ_8, λ_9 ni chiqaring. Farqni izohlang.
20. **Autoencoder.** AE(k): encoder `Linear(64,64)-ReLU-Linear(64,k)`, decoder `Linear(k,64)-ReLU-Linear(64,64)-Sigmoid`. `train_ae(X, k, epochs=60)`: `torch.manual_seed(42)`, Adam $3e-3$, MSE, 64 lik paketlar (`randperm`, Generator seed 42). $k = 2, 8$: test MSE va PSNR; bazaviy usul — train o'rtacha rasmi.
21. **Anomaliyani aniqlash.** `train_ae` ni faqat `ytr != 8` rasmlarda ($k = 8$) o'qiting. Har test rasmi uchun o'rtacha kvadratik tiklash xatosi; "8" — anomaliya. `roc_auc_score`; chegara — normal train xatolarining 95%-kvantili: aniqlangan 8 lar ulushi va soxta signal ulushi.
22. **Denoising va Tweedie.** `torch.manual_seed(42)`; `Linear(1,32)-Tanh-Linear(32,1)`; Adam $1e-2$, 2000 qadam; har qadamda $\mathbf{x} = 2 + \text{randn}(512, 1)$, $\mathbf{xt} = \mathbf{x} + \sqrt{0.5} * \text{randn}(512, 1)$, `MSE(net(xt), x)`. $\tilde{x} \in \{0,5; 2; 3,5\}$ da natijani 2-topshiriq formulasi bilan solishtiring.
23. **VAE (load_digits).** VAE(k=8, h=128): `Linear(64,128)-ReLU` → μ , $\logvar$ (`Linear(128,8)`); decoder `Linear(8,128)-ReLU-Linear(128,64)` (logitlar). `train_vae(beta, k=8, epochs=80)`: `torch.manual_seed(42)`, Adam $2e-3$, paket 64 (`randperm`, seed 42), yo'qotish `vae_loss(...)/len(batch)`. Baholash (`torch.manual_seed(0)`, 10 ta ϵ): test tiklash (BCE yig'indisi/rasm), har o'lcham KL, -ELBO, faol o'lchamlar ($KL > 0,01$).
24. β ta'siri. $\beta \in \{0, 1, 4\}$ uchun tiklash, KL va faol o'lchamlar jadvali; posterior collapse ni aniqlang.
25. **Namunalar sifati: AE va VAE.** `LogisticRegression(max_iter=3000)` ni train piksellarida o'qiting (test aniqligi). $\mathbf{z} = \text{torch.randn}(450, 8, \text{generator=Generator().manual_seed(7)})$ ni AE (20, $k = 8$) va VAE ($\beta = 1$, `sigmoid`) decoderlariga bering. Namunalar va test rasmlari uchun: piksel std (namunalar bo'yicha, piksellar o'rtachasi), o'rtacha $\max p(y|\mathbf{x})$, sinflar histogrammasi.

26. **1D GAN.** Haqiqiy $\mathcal{N}(4; 1, 25^2)$; `torch.manual_seed(42)`; G: `Linear(1,16)-ReLU-Linear(16,1)`; D: `Linear(1,32)-LeakyReLU(0.2)-Linear(32,32)-LeakyReLU(0.2)-Linear(32,1)` (logit); Adam $1e-3$, `betas=(0.5,0.999)`; 3000 qadam, paket 256, non-saturating, detach. L_D , L_G , $G(z)$ o'rtacha/std (10 000 namuna), $D(x)$, $D(G(z))$.
27. **Bir vaqtda va navbatma-navbat.** 13-topshiriqdagi o'yinni 100 qadam simulyatsiya qiling: bir vaqtda va navbatma-navbat ($x \leftarrow x - \eta y$, keyin yangi x bilan $y \leftarrow y + \eta x$); 10 va 100 qadamdagi normalar.
28. **D^* ni o'rganish.** Logistik diskriminator $D(x) = \sigma(ax + b)$, $a = b = 0$; `torch.manual_seed(42)`; Adam 0.05, 1500 qadam; har qadamda $x_r = \text{randn}(512)$, $x_g = 2 + \text{randn}(512)$; BCE. (a, b) ni va $x = 0, 1, 2$ da D ni 10-topshiriq bilan solishtiring.
29. **GAN (load_digits) + FID.** $k = 16$; G: `Linear(16,128)-ReLU-Linear(128,64)-Sigmoid`; D: `Linear(64,128)-LeakyReLU(0.2)-Linear(128,1)`; `torch.manual_seed(42)`; Adam $1e-3$, `betas=(0.5,0.999)`; 4000 qadam; har qadamda 64 ta train rasmi (`randint`, Generator seed 42) va `torch.randn(64,16)`. L_D , L_G , test va namunalar uchun D (namunalar: `randn(450,16, generator=seed 7)`), namunalar sinflari (25-topshiriq klassifikatori). `fid(A, B)` (`scipy.linalg.sqrtm(...).real`); avval 1D: `default_rng(1).normal(0,1,(200000,1))` va `default_rng(2).normal(0.5,0.5,(200000,1))`. So'ng PCA(20, `random_state=42`) (train'da) belgilari bo'yicha FID(test, .): train, VAE, GAN, tekis shovqin `default_rng(42).random((450,64))`.
30. **Normalizing flow (RealNVP).** `make_moons(n_samples=3000, noise=0.05, random_state=42)`, standartlashtirish; birinchi 2000 — train, qolgani — test. Affine coupling: bir koordinata a dan `Linear(1,64)-ReLU-Linear(64,64)-ReLU-Linear(64,2)` $\rightarrow (s, t)$, $s \leftarrow \tanh s$, $b \mapsto be^s + t$, $\log|\det| = s$; 6 qatlam, navbatma-navbat almashadi. `torch.manual_seed(42)`; Adam $2e-3$; 2000 qadam, paket 256 (`randint`, seed 42); yo'qotish $-\log p(x)$. Test o'rtacha $\log p$ ni to'liq kovariatsiyali Gauss bilan solishtiring; teskarilash xatosi; parametrlar; `randn(2000,2, generator=seed 1)` dan namunalar uchun eng yaqin train nuqtagacha median masofa va $> 0,3$ ulushi.